

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Chest tightness										
Asthma Symptoms, Shortness of breath	Lewis et al.	2013	Detroit, MI	5-12		D8HourMax	0.00423	0.00386	Logistic	All year
Asthma Onset	Tétreault et al.	2016	Québec, Canada	0-17		Annual (D8HourMax)	0.02075	0.00146	Log-linear	Warm season; separate HIFs for ages 0-4; 5-17

F.5.1 Lewis et al. (2013)

Lewis et al. (2013) studied the effects of short-term O₃ exposure on frequency of asthma symptoms in an asthmatic population of primarily lower-income, African American and Latino children (aged five-12 years) in East and Southwest Detroit, MI. Authors obtained health and population subgroup data through questionnaires filled out by parents or guardians for 14 consecutive days in each studied season. Questionnaires highlighted participant's asthma symptoms (cough, wheeze, shortness of breath, chest tightness), population subgroup information, medication use, and presence of second-hand smoke. The authors acquired maximum one-hour and maximum 8-hour O₃ concentrations and meteorological data from two community-level monitors placed on East and Southwest Detroit, MI school rooftops. Lewis et al. (2013) implemented a combination of generalized estimating equations and alternative logistic regression models to estimate the associations between short-term O₃ exposure and rate of asthma symptoms. Models adjusted for age, sex, location (Eastside or Southwest), race, household income, smoker in the home, season, and variables for companion home intervention study (control or intervention), time (pre- or post-intervention), and the interaction between intervention group status and time. Lewis et al. (2013) observed positive associations between short-term O₃ exposure and asthma symptoms.

Asthma Symptoms

In single-pollutant models, the coefficient and standard error are based on the all year odds ratios of 1.12 (95% CI: 0.99-1.25) for cough, 1.13 (95% CI: 0.99-1.28) for wheeze, 1.20 (95% CI: 1.02-1.40) for chest tightness, and 1.07 (95% CI: 0.95-1.21) for shortness of breath, all for a 16 ppb (interquartile range) increase in 8-hour maximum O₃ concentrations (five-day average lag) (Lewis et al. 2013, Figure 1C).

F.5.2 Tétreault et al. (2016)

Tétreault et al. (2016) investigated the effects of long-term O₃ exposure on asthma onset in children (aged zero-12 years) from Québec, Canada. The study followed participants from the Québec Integrated Chronic Disease Surveillance System open birth cohort between 1999 and 2011. The authors defined new cases of asthma based

on hospital discharge reports and physician diagnoses (two diagnoses within a two-year span). Monitor data (Canadian National Air Pollution Surveillance network) and land-use mixed effect models estimated warm season (June to August) O₃ exposures. Authors assessed associations with asthma onset with both time of birth and time-varying exposure models and adjusted for year of birth, sex, and indices of social and material deprivation. Tétreault et al. (2016) used Cox proportional hazard models to observe associations between long-term O₃ exposure and asthma onset in children.

As the physiology and etiology of lung development in children is similar in children 6-17 (Baena-Cagnani et al., 2007, Guerra et al., 2004, Ochs et al., 2004, Sparrow et al., 1991, Trivedi and Denton, 2019), we apply the 4-12 year age-stratified effect estimate from Tétreault et al. (2016) to children ages 4-17.

Incidence, Asthma

In a single-pollutant time-varying model, the coefficient and standard error were estimated from a warm-season hazard ratio of 1.07 (95% CI: 1.06-1.08) for a 3.26 ppb (interquartile range) increase in annual O₃ concentrations (Tétreault et al. 2016, Table 5).

F.6 Sensitivity Analysis – General

Table F-6 summarizes the ozone health impact functions considered by EPA to be sensitivity analyses. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table F-6. Core Health Impact Functions for Ozone Sensitivity Analyses

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	PM _{2.5}	Annual (D8HourMax)	0.001094	0.000050	Log-linear	All Cause, warm season
Incidence, Asthma	Garcia et al.	2019	12 Southern California communities	9-18		Annual (D8HourMax)	0.016946	0.010941	Log-linear	All year
Mortality, Respiratory	Katsouyanni et al.	2009	90 U.S. Cities	0-99	PM ₁₀	D1HourMax	0.001921	0.001297	Log-linear	Multi-pollutant, lag 1, warm season
Mortality, Respiratory	Katsouyanni et al.	2009	90 U.S. Cities	0-99	PM ₁₀	D8HourMax	0.001921	0.001297	Log-linear	Multi-pollutant, lag 1, warm season, 8-hour max from 1-hour max using adjustment factor of 1.13, resulting in effective beta of 0.002175.
Mortality, Respiratory	Katsouyanni et al.	2009	90 U.S. Cities	0-99		D1HourMax	0.0015	0.000593	Log-linear	Single-pollutant, lag 1, warm season
Mortality, Respiratory	Katsouyanni et al.	2009	90 U.S. Cities	0-99		D8HourMax	0.0015	0.000593	Log-linear	Single-pollutant, lag-1, warm season, 8-hour max from 1-hour max using adjustment factor of 1.13, resulting in effective beta of

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
										0.001695.
Mortality, All Cause	Turner et al.	2016	Nationwide	30-99	PM _{2.5}	Annual (D8HourMax)	0.001980	0.000500	Log-linear	All Cause, warm season, multi-pollutant
Mortality, Respiratory	Turner et al.	2016	Nationwide	30-99		Annual (D8HourMax)	0.013103	0.001791	Log-linear	Single-pollutant
Mortality, Respiratory	Turner et al.	2016	Nationwide	30-99	PM _{2.5}	Annual (D8HourMax)	0.011333	0.001823	Log-linear	Multi-pollutant

F.6.1 Di et al. (2017)

Di et al. (2017) evaluated the relationship between long-term ozone exposure and all-cause mortality in nearly 61 million U.S. Medicare enrollees (over the age of 64) through 460 million person-years of follow-up and roughly 22 million observed deaths. This cohort comprised approximately 15% of the total U.S. population, included people living in rural areas, and is one of the largest cohort studies published to date. The authors modeled warm season ozone exposure across the contiguous U.S. using a hybrid methodology that included land use regression, satellite data, and monitor data, and resolved estimations to 1 x 1-kilometer areas. Di et al. (2017) used two-pollutant Cox proportional-hazards models with a generalized estimating equation. The authors controlled for population subgroup characteristics, Medicaid eligibility, and area-level covariates.

All-Cause Mortality

In a two-pollutant model, the coefficient and standard error for ozone are estimated from the hazard ratio (1.011) and 95% confidence interval of (1.010-1.012) associated with a change in annual mean ozone exposure of 10.0 ppb (Di et al., 2017, Table 2 Main Analysis).

F.6.2 Garcia et al. (2019)

Garcia et al. (2019) examined the associations between long-term ozone exposure and asthma onset in children (aged nine-18 years) with no history of asthma in Southern California. The authors followed three waves of participants from the Children's Health Study for eight years between 1993 and 2014. Garcia et al. (2019) obtained health and population subgroup data from parents, guardians, or participants, who completed questionnaires annually. In order to calculate annual mean, community-level ozone exposure, the authors acquired daily eight-hour mean ozone concentrations through ambient air pollution monitors. Multi-level Poisson regression models with one-year lag showed no statistically significant associations between long-term ozone exposure and asthma onset in children. Models adjusted for population subgroup variables as well as factors pertaining to family medical history, environmental factors, and near-roadway pollution.

Incidence, Asthma

In a single-pollutant, all year model, the coefficient and standard error were estimated from an incidence rate ratio of 0.86 (95% CI: 0.71-1.04) for an 8.9 ppb decrease in eight-hour mean ozone concentrations (Garcia et al. 2019, Table 2). For consistency with the other HIFs, we convert this to a rate ratio for an increase in ozone, by taking the inverse of the reported incidence rate ratio, giving a rate ratio of 1.163 (95% CI: 0.962-1.408) for an 8.9 ppb increase in eight-hour mean ozone concentrations.

F.6.3 Katsouyanni et al. (2009)

See full study description under Katsouyanni et al. (2009) in Appendix F, Section F.1.1.

Short-term Mortality, Respiratory (Multi-Pollutant)

In a multi-pollutant model, the coefficient and standard error are based on the summer-only penalized spline estimate of respiratory mortality of 0.99% (-0.33, 2.31%) per 10 $\mu\text{g}/\text{m}^3$ (equivalent to 5.1 ppb) increase in O_3 lagged by 1 day (Katsouyanni et al. 2009, Table 24: Lag 1; Penalized splines; Controlling for PM_{10}).

The Health Impact Function was adjusted from the daily 1-hour max metric to the daily 8-hour max metric using a ratio of 1.13 (ratio of 1-hour max to 8-hour max ozone) (Anderson and Bell 2010, Table 2).

Short-term Mortality, Respiratory (Single-Pollutant)

In a single-pollutant model, the coefficient and standard error are based on the summer-only penalized spline estimate of respiratory mortality of 0.77% (0.17, 1.37%) per 10 $\mu\text{g}/\text{m}^3$ (equivalent to 5.1 ppb) increase in O_3 lagged by 1 day (Katsouyanni et al. 2009, Table 24: Lag 1; Penalized splines; O_3 Results).

The Health Impact Function was adjusted from the daily 1-hour max metric to the daily 8-hour max metric using a ratio of 1.13 (ratio of 1-hour max to 8-hour max ozone) (Anderson and Bell 2010, Table 2).

F.6.4 Turner et al. (2016)

See full study description under Turner et al. (2016) in Appendix F, Section F.1.2.

Long-term Mortality, All Cause

In a multi-pollutant model, the coefficient and standard error are based on the warm-season specific hazard ratio of 1.02 (1.01-1.03) per 10 ppb increase in seasonal average of daily 8-hour maximum O_3 concentrations (Turner et al. 2016, Table E9: HBM O_3 , multipollutant model data, fully adjusted).